

Collatz-Conjecture Proved and Halemane-Conjecture Proposed

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Abstract

This paper presents a definitive proof of the Collatz Conjecture, that is also known as the Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture, asserting the convergence of the Collatz sequence to the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$.

An algebraic-system-structure that is an exact mathematical model to represent the Collatz $(3x+1)$ system; is designed similar to a bounded finite small-sized ideal-based graded-algebraic-filtration-structure defined on a modulo-3 quotient-semiring generated by the primitive root 2. The trivial-cycle of the Collatz-map is bypassed through an initialization-phase for this algebraic-system-structure, starting directly with a 3-layered structure with the modulo-9 coprime-layer as its topmost layer. This also facilitates a clear distinction among the six different possible

combinations of the modulo-3 residue-classes and the modulo-6 residue-classes that are associated with any positive odd number; which is blurred in smaller structures. The Collatz odd-operator $(3x+1)$ when x is a positive odd number; acts as a filtration shifting global-affine-transformation; achieving a Euclidean expansion shift to the topmost coprime-layer of that filtration structure; while also avoiding the modulo multiple layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent elements (dead-end zero-divisors).

Similarly, an algebraic-system-structure that is an exact mathematical model to represent the inverse Collatz $(3x+1)$ system; is designed similar to a bounded finite small-sized ideal-based graded-algebraic-filtration-structure defined on a modulo-2 quotient-semiring generated by the pair of roots -1 and 3. The trivial-cycle of the Collatz-map is bypassed through an initialization-phase for this algebraic-system-structure, starting directly with a 5-layered structure with the modulo-16 coprime-layer as its topmost layer. This also facilitates a clear distinction among the six different possible combinations of the modulo-4 residue-classes and the modulo-6 residue-classes that are associated with any positive odd number; which is blurred in smaller structures. A global-affine-transformation, defined by the condensed compact inverse Collatz odd-to-odd (CTUHSK) function; works solely at the topmost coprime-layer of this filtration structure; avoiding the modulo-multiple-layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent-elements (dead-end zero-divisors).

This algebraic system design establishes that the Collatz-Map covers all the relevant (modulo-3 & modulo-6) modular-residue-classes and also covers all the possible valid triplet-combinations of (1) input-values (2) operations and (3)

output-values; governed by the modular-periodicity characteristic (resulting in the self-similar symmetry structure) of the Collatz-Map.

Also, this algebraic system design establishes that the transitive closure of the subset $\{1,5,3\}$ under the condensed compact inverse Collatz odd-to-odd (CTUHSK) function is the entire set of all positive odd integers; asserting that the Collatz sequence starting from any given positive integer converges to the trivial-cycle.

A deterministic discrete finite-state autonomous system (DDFSAS) model for the Collatz system leads to the Halemane-Assertion that there exist exactly 108 distinctly different possible classes of odd $(3x+1)$ operations in the Collatz system.

Halemane-Conjecture states that the maximum number of odd $(3x+1)$ operations required to reach the trivial-cycle $\{1 \Leftarrow 2 \Leftarrow 4\}$; starting from any given positive integer greater than one and moving along the Collatz sequence; is limited by that given number itself; the triad $\{31 \Leftarrow 41 \Leftarrow 27\}$ is an exceptional limiting case.

Keywords: Binary-Exponential-Ladder; Modular-Periodicity;

Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) System;

Deterministic Discrete Finite-State Autonomous System (DDFSAS) Model;

Halemane-Assertion; Halemane-Conjecture;

Halemane's Theory of the CTUHSK System.

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1. Introduction

The Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture, simply referred to as the Collatz Conjecture [§1] states that the Collatz Sequence, also referred to as the Collatz $(3x+1)$ Sequence, converges to the trivial-cycle, that is, $\{(1 \Leftarrow 2 \Leftarrow 4)\}$, starting from any positive integer.

The Collatz $(3x+1)$ sequence, starting with any given positive integer is the sequence of numbers obtained by the repeated application of the Collatz function defined on the set of positive integers, given by [Eqn.1] as –

$$\begin{aligned} \text{if } (x \equiv 0 \pmod{2}) \text{ then } S(x) &:= S_0(x) := (x/2); \\ \text{elseif } (x \equiv 1 \pmod{2}) \text{ then } S(x) &:= S_1(x) := (3x+1); \end{aligned} \quad [\text{Eqn.1}]$$

Note that the Collatz successor function $S(x)$ is a unique one-to-one mapping.

Lagarias [§3]&[§4] gives an exhaustive annotated bibliography, whereas Lagarias [§5]&[§6] gives an elaborate overview of the Collatz Problem, also referred to as the “ $3x+1$ problem”, referring to it as “The Ultimate Challenge”. Guy [§2] has been somewhat pessimistic in advising researchers “Don’t Try to Solve These Problems”. Lagarius [§5]&[§6] refers to the statement by the most revered Mathematician and Number Theory Expert Paul Erdos, who said that - "Mathematics is not yet ready for such problems" - about the Collatz Conjecture. The Collatz $(3x+1)$ Sequence is indeed a highly complex chaotic system defined or generated with one of the simplest deterministic well-defined equations, that too, just two in number. We will not get into any more of the reported literature except the problem definition;

but will simply focus on a rather shockingly simple approach to solve this amazing long-standing unresolved problem.

This paper presents an algebraic system model for the Collatz $(3x+1)$ system; asserting the convergence of the Collatz sequence to the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$. Also, a deterministic discrete finite-state autonomous system (DDFSAS) model is presented, having 108 distinct classes of Collatz (odd-to-odd) operations.

2. Algebraic System Structure for the Collatz Function

An exact mathematical representation of the Collatz function is achieved through the design of an algebraic-system-structure similar to a bounded finite small-sized ideal-based graded-algebraic-filtration-structure defined on a modulo-3 quotient-semiring, generated by the primitive root 2; having the following characteristics.

2.1 BASE SEMIRING AND IDEAL

Start with the semiring of non-negative integers, $R = (\mathbb{Z}_{\geq 0}, +, \cdot)$;

The modulo-3 quotient semiring is constructed by considering the ideal $I = \langle 3 \rangle$ defined by the modulo-base 3.

2.2 PRIMITIVE-ROOT

The element 2 serves as a primitive-root modulo-3, because $2^2 \equiv 1 \pmod{3}$. Every division by this primitive-root 2 (whenever the number is even, as per the Collatz

function) is achieved through multiplication by the modular inverse of this primitive-root in the specific quotient-semiring; for example, through a multiplication by 2 in modulo-3 semiring, whereas it is through a multiplication by 5 in modulo-9 semiring, etc.

2.3 DIVISIBILITY-CLASSES and IDEAL-BASED FILTRATION-STRUCTURE

The divisibility-classes based on the modulo-3 ideal-base (used for the creation of the quotient-semiring) defines the subring layers in the filtration-structure. To start with, in the most trivial-case, we have a modulo-multiplier layer (modulo- 3^0 layer or zero-layer) at the bottom and also a modulo- 3^1 layer (layer-1) as the topmost coprime-layer. This is considered as a trivial one-layer algebraic-filtration-structure $F(3, 2; 1)$. This trivial filtration-structure $F(3, 2; 1)$ by itself is indeed an exact representation of the entire Collatz System Dynamics or the entire Collatz-Map.

The primitive-root 2 generates the multiplicative structure within any given subring; and the elements transition down the filtration hierarchy by exact division by the ideal-base 3, although the Collatz function does not have any explicit operation of division by 3 and therefore we do not explicitly use this feature. However, note that when we refine the trivial structure $F(3,2;1)$ by expansion; introducing a modulo-9 coprime layer as layer-2 to construct $F(3,2;2)$; the elements $[1]_9$ $[4]_9$ and $[7]_9$ of $S(3,2;2;2)$ together form a refinement by expansion of the earlier element $[1]_3$ in $S(3,2;1;1)$ of the trivial structure $F(3,2;1)$; whereas $S(3,2;2;1)$ that is, the new layer-1 of this newly constructed $F(3,2;2)$ has its elements updated by $3 \times \{S(3,2;1;1)\}$ that is, $3 \times \{[1]_3; [2]_3\}$. That shows the implicit result of the exact division by the ideal-base 3 at every step as we move down the tower.

Note that the process of repeated division by the primitive-root 2, within the subring of any given filtration layer, produces two partitions of the subring consisting of alternating elements of the orbit in the subring; one belonging to the modulo-3 residue-class $[1]_3$ and the other belonging to the modulo-3 residue class $[2]_3$; of almost equal-size; the relative size breakup alternates between these two residue classes as we move up or down the filtration tower.

2.4 INITIALIZATION PHASE TO BYPASS THE TRIVIAL CYCLE

The trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ can be effectively bypassed by an appropriate initialization-phase for this filtration structure, starting directly with the modul-9 coprime-layer; that is, $F(3,2;2)$.

2.5 EUCLIDEAN EXPANSION SHIFT TO THE TOPMOST COPRIME LAYER

The Collatz odd-operator $S_1(x) := (3x+1)$ when x is a positive odd number; acts as a filtration shifting global-affine-transformation; achieving a Euclidean expansion shift to the topmost coprime-layer of that filtration structure; while also avoiding the modulo multiple layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent elements (dead-end zero-divisors) like $(6m-3)$ and $(6m-0)$. The output of this affine transformation results in a number that is always a positive even number of the form $(6m-2)$ that belongs to the modulo-3 residue class $[1]_3$ at the topmost coprime-layer.

3. The Inverse Collatz Function

The inverse of the condensed compact Collatz odd-to-odd (CTUHSK) function; $P(x)$, defined on the set of positive odd numbers, is given by:

$$\begin{aligned} \text{if } (n \equiv 1 \pmod{6}) \text{ then } \{P(n)\} &:= \{P_1(n, w) := (n \cdot 2^w - 1)/3\}; \\ \text{elseif } (m \equiv 5 \pmod{6}) \text{ then } \{P(m)\} &:= \{P_5(m, v) := (m \cdot 2^v - 1)/3\}; \end{aligned} \quad [\text{Eqn.2}]$$

Note that the Collatz predecessor function $\{P(x)\}$ is a one-to-many mapping.

Here, binary exponent w is a positive even integer and the binary exponent v is a positive odd integer; and $\{P(x)\}$ correspond to the set of all immediate-predecessors of x in the condensed compact inverse Collatz odd-to-odd CTUHSK sequence (reverse of the Collatz sequence, wherein the intervening nodes corresponding to the positive even integers in the standard canonical Collatz sequence are suppressed or rather absorbed into the immediate predecessor node).

4. Modular-Periodicity of the Collatz $(3x+1)$ System

Observe that the immediate-predecessor relation given by $P_1(n, w)$ and $P_5(m, v)$ in [Eqn.2] above satisfies the modular-periodicity property (characteristic of the CTUHSK system; governing the self-similar symmetry in the arborescence structure) of the Collatz-Map; given by [Eqn.3] here below.

$$[P_1((6k+1), (2j))]_3 = [j]_3 - [k]_3;$$

and

$$[P_5((6k+5), (2j-1))]_3 = [j]_3 - 1 + [k]_3; \quad [\text{Eqn.3}]$$

Note that [Eqn.3] describes the exceptionally unique modular (modulo-3 and modulo-6) periodicity relation among the three parameters defining the predecessor relation, as represented by the inverse Collatz function given in [Eqn.2] above; namely, (1) the input value (2) the binary exponent value and (3) the output value.

5. Algebraic System Structure for the Inverse Collatz Function

An Algebraic System Structure is designed that provides an exact representation of the condensed compact inverse Collatz odd-to-odd (CTUHSK) function with the following characteristics.

5.1 BASE SEMIRING AND IDEAL

Start with the semiring of non-negative integers, $R = (\mathbb{Z}_{\geq 0}, +, \cdot)$;

The modulo-2 quotient semiring is constructed by considering the ideal $I = \langle 2 \rangle$ defined by the modulo-base 2.

5.2 PAIR OF ROOTS AS GENERATORS

The modulo- 2^k quotient semiring, for any $k > 1$, has -1 and 3 as the pair of roots serve as generators; creating two distinct orbits (of exactly equal size) one being a mirror image of the other with respect to the modulus-base 2^k . The division-by-3 operation (as required by the inverse Collatz function) within any the semiring

structure, is achieved through multiplication by the modular inverse of 3 in the specific quotient-semiring; for example, multiplying by 11 in modulo-16 semiring.

5.3 INITIALIZATION PHASE TO BYPASS THE TRIVIAL CYCLE

The trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ can be bypassed by an initialization-phase for this filtration structure, starting directly with the modul-16 coprime-layer. This enables the clear distinction among the different modular (modulo-4 & modulo-6) residue classes of the various parameters in the problem.

5.4 DIVISIBILITY-CLASSES

The divisibility-classes based on the modulo-2 base (used for the creation of the quotient semiring) defines the subring layers in the filtration-structure. The quotient semirings defined with respect to modulus base 2^k characterize the layer- k with divisibility-order k in this filtration structure.

5.5 IDEAL-BASED FILTRATION STRUCTURE

To define a hierarchical filtration structure, we create a tower of subrings or ideals, corresponding to the divisibility-order with respect to the modulo-base 2; the bottommost being the zero-layer or the all-absorbing-layer. The root 3 generates the multiplicative structure (corresponding to the division-by-3-operation in the inverse Collatz function) within any given subring; and the elements transition down the filtration hierarchy by exact division by 2; although the inverse Collatz function does not have any operation of division by 2 (similar to the situation of exact division by 3 seen in the earlier structure).

5.6 TRANSFORMATIONS AT THE TOPMOST COPRIME LAYER

The inverse Collatz operators $P(x)$ in [Eqn.2] above, suggest the use of a transformations $f(x) := (2^w \cdot x - 1)/3$ and/or $f(x) := (2^v \cdot x - 1)/3$; working at the topmost coprime-layer; avoiding the modulo-multiple-layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent-elements (dead-end zero-divisors) like $(4m-2)$ and $(4m-0)$.

6. An Elegant Proof of the Collatz Conjecture

Observe that both the domain as well as the range of the condensed compact inverse Collatz odd-to-odd (CTUHSK) function (a one-to-many mapping); given in [Eqn.2] above; being the set of all positive odd integers; covers every relevant modular (modulo-3 & modulo-6) residue-class.

Also, observe that by the very design of the algebraic-system-structure, the transitive closure of the subset $\{1,5,3\}$ under the condensed compact inverse Collatz odd-to-odd function is the entire set of all positive odd numbers (neither missing any nor including any extraneous-elements); covering all the relevant (modulo-3 and modulo-6) modular-residue-classes; also covering all the possible valid triplet-combinations of (1) input-values (2) operations and (3) output-values; governed by the exceptionally unique modular-periodicity characteristic requirements/constraints as presented in [Eqn.3]; also resulting in the self-similar symmetry structure of the Collatz-Map. This is a definitive proof of the Collatz conjecture.

7. Deterministic Discrete Finite-State Autonomous System Model

The Collatz $(3x+1)$ system is a deterministic discrete finite-state autonomous system (DDFSAS). The set of distinct possible system state transitions in this DDFSAS is uniquely defined and identified by a combination of the appropriate modular-residue-classes of the five parameter-classes that govern the condensed compact inverse Collatz odd-to-odd (CTUHSK) function; $P(x)$ as in [Eqn.2] above which itself has two distinct operators $P_1(n, w)$ and $P_5(m, v)$ for the generic function $P(x)$.

The first parameter-class is the modulo-3 residue class associated with the quotient in the modulo-6 coset-representative of the positive odd number input argument value x to $P(x)$; that is, $k(\text{MOD}3)$ where $x=6k+x(\text{MOD}6)$. The second parameter-class is the modulo-6 residue-class of the positive odd number input argument value x to $P(x)$; that is, exactly one of the two possibilities, either $[1]_6$ or $[5]_6$; associated with the corresponding operators $P_1(n, w)$ and $P_5(m, v)$. The third parameter-class is the modulo-6 residue-class of the appropriate valid binary exponent (positive integer) w in $P_1(n, w)$ or v in $P_5(m, v)$; which itself is solely dependent on the first parameter, that is, from exactly one of the two sets of exponents, either $\{[1]_6; [3]_6; [5]_6\}$ or $\{[0]_6; [2]_6; [4]_6\}$. The fourth and fifth parameter-classes correspond to the output value of $P(x)$ exactly similar to the first and second parameter-classes corresponding to the input value. Therefore, the number of distinct possible combinations of these five discrete finite sets of parameter-classes is $3 \times 2 \times 3 \times 3 \times 2 = 108$.

8. Halemane-Assertion and Halemane-Conjecture

Halemane-Assertion states that there exist exactly 108 distinctly different possible classes of odd $(3x+1)$ operations in the Collatz system, arising from the exceptionally unique modular (modulo-3 and modulo-6) periodicity characteristic (also resulting in the self-similar symmetry structure) of the Collatz system.

Halemane-Conjecture [§7] states that the upper bound on the number of odd $(3x+1)$ operations required to reach the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$; starting from any given positive integer greater than one and moving along the Collatz sequence; is the given number itself; the triad $\{(31 \Leftarrow 41 \Leftarrow 27)\}$ is an exceptional limiting case.

9. Summary

An exact mathematical model of the condensed compact inverse Collatz odd-to-odd system is achieved through an algebraic-system-structure that is designed similar to a bounded finite ideal-based graded-algebraic-filtration-structure defined on a modulo-2 quotient-semiring generated by the pair of roots -1 and 3. By this very design it is observed that the transitive closure of the subset $\{1,5,3\}$ under the inverse Collatz function is the entire set of all positive integers; providing a definitive proof of the Collatz conjecture.

The Halemane-Assertion that there exist exactly 108 distinctly different possible classes of odd $(3x+1)$ operations in the Collatz system, is directly linked to the design of a deterministic discrete finite-state autonomous system (DDFSAS) model for the Collatz system.

Halemane-Conjecture [§7] states that the upper bound on the number of Collatz odd $(3x+1)$ operations required to reach the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$; starting from any natural number greater than one and moving along the Collatz sequence; is given number itself; the triad $\{(31 \Leftarrow 41 \Leftarrow 27)\}$ is an exceptional limiting case.

9. Recommended Reading

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10. Declaration Regarding Affiliation and Funding

I, Dr(Prof) Keshava Prasad Halemane, hereby declare that I am a Professor retired as on 2017JAN31 from National Institute of Technology Karnataka Surathkal India, and I am not affiliated to any institution or organization or corporation or any other agency or whatever. This research work has been conducted entirely by me on my own as an Independent Researcher, and that I have not received any funding from any source other than my own savings, and I do not have any obligations or encumbrances of any kind, neither financial nor legal nor of any other kind, regarding the contents of the manuscript - of which I am the original author and creator. Also, I hereby declare there is no conflict of interests.

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12. Dedication

To my ಅಜ್ಜ(ajja) Karinja Halemane Keshava Bhat & ಅಜ್ಜಿ(ajji) Thirumaleshwari, ಅಪ್ಪ(appa) Shama Bhat & ಅಮ್ಮ(amma) Thirumaleshwari, for their *teachings through love, that quality matters more than quantity*; to my wife Vijayalakshmi for her *ever consistent love & support*; to my daughter [Sriwidya.Bharati](#) and my twin sons [Sriwidya.Ramana](#) & [Sriwidya.Prawina](#) for their *love & affection*.